Semantic-Structural Decomposition for Uncertainty in Knowledge Graph Embeddings

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Abstract

Knowledge graph embedding methods increasingly incorporate uncertainty quantification, yet their reliability for out-of-distribution detection remains poorly understood. This paper identifies a fundamental limitation in probabilistic approaches: learned entity variances are relation-agnostic and cannot capture whether an entity has been observed with a specific relation. This observation motivates a *semantic-structural decomposition*, where semantic uncertainty reflects embedding quality and structural uncertainty reflects observation patterns. Coverage—a simple relation-specific lookup—provides a sufficient statistic for structural uncertainty with a closed-form AUROC bound (validated within 4% error). Neither component alone suffices: their combination achieves 0.87–0.96 AUROC across three benchmarks with 14–32% improvement over the best single signal. This analysis reveals why existing methods struggle and provides a principled framework for uncertainty quantification in knowledge graphs.

1 Introduction

Knowledge graphs power critical applications: drug interaction prediction, fraud detection, recommendation systems. When these systems make predictions about unseen entities or novel relationships, knowing *when to trust the output* is essential for safe deployment. This motivates out-of-distribution (OOD) detection—distinguishing reliable predictions from uncertain ones.

Probabilistic knowledge graph embedding methods address this by learning distributions over entity embeddings rather than point estimates. GP-KGE [Chen et al., 2021b] learns a variance σ_e^2 for each entity, capturing how "uncertain" the model is about that entity's representation. The intuition is appealing: entities seen frequently should have low variance (well-constrained embeddings), while rare entities should have high variance.

A fundamental limitation. This intuition breaks down because the learned variance is *relation-agnostic*. Consider an entity e that appears frequently with relation r_1 (e.g., `born_in`) but never with relation r_2 (e.g., `works_at`). GP-KGE assigns a single variance σ_e^2 to entity e, which is the same for both relations. But the model's uncertainty about $(e, r_1, ?)$ and $(e, r_2, ?)$ should differ—e has been observed with r_1 but not with r_2.

This observation leads to the central insight: *uncertainty in knowledge graphs decomposes into semantic and structural components, and neither alone suffices for OOD detection.* **Semantic uncertainty** (GP variance) captures embedding quality—rare entities have high variance. **Structural uncertainty** (coverage) captures whether a specific entity-relation pair has been observed: $c(e, r) = 1$ if entity e appears with relation r in training. Coverage is relation-specific by construction, addressing GP's limitation.

Contributions.

1. A fundamental limitation in probabilistic KGE: learned variances are relation-agnostic and miss structural uncertainty.

2. A semantic-structural decomposition framework that separates these orthogonal uncertainty types.

3. A closed-form AUROC bound for coverage-based detection, validated empirically with <4% error.

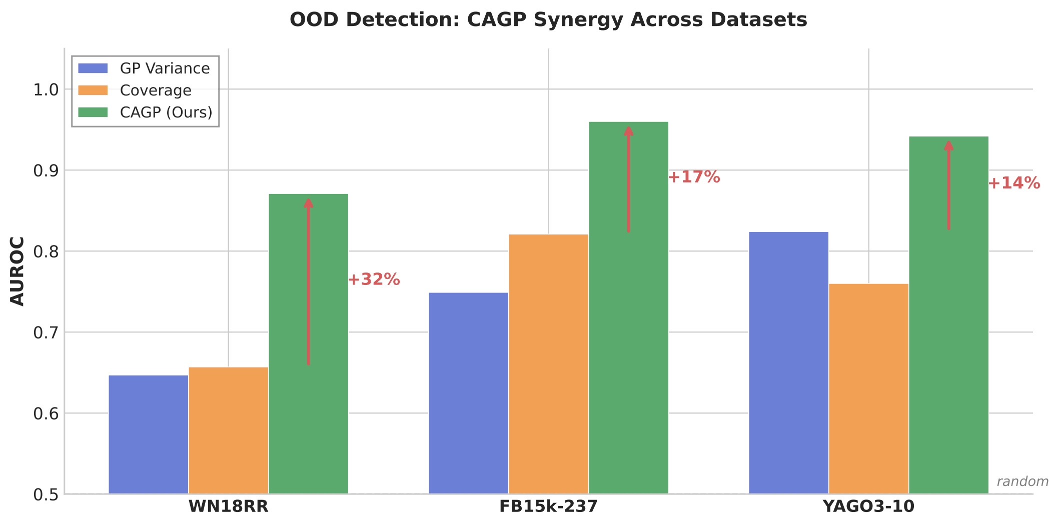

Figure 1: OOD detection AUROC. CAGP consistently outperforms single signals with 14–32% synergy.

4. Empirical demonstration that combining both signals (CAGP) achieves 0.87–0.96 AUROC with 14–32% synergy across three benchmarks.

The key finding is not the CAGP algorithm itself—which is a simple linear combination—but the *insight* that probabilistic KGE methods have a structural blind spot that cannot be learned away. Figure 1 summarizes the main results.

2 Related Work

Uncertainty in Knowledge Graph Embeddings. UKGE [Chen et al., 2019] associates confidence scores with triples, reflecting triple-level reliability rather than entity-level uncertainty. BEUrRE [Chen et al., 2021a] uses box embeddings where box size indicates uncertainty, but uncertainty remains entity-level and relation-agnostic. GP-KGE learns Gaussian distributions over entity embeddings. The limitation identified in this paper—that learned variances cannot capture relation-specific uncertainty—applies to all these methods.

OOD Detection in Deep Learning. MC Dropout [Gal and Ghahramani, 2016] estimates uncertainty via dropout at test time. Deep Ensembles [Lakshminarayanan et al., 2017] aggregate predictions from independently trained models. These methods are relation-agnostic when applied to KG embeddings: they capture model uncertainty but not structural observation patterns.

Graph-Based Uncertainty. Graph Posterior Networks [Stadler et al., 2021] propagate uncertainty through graph structure for node classification. GGPN [Chen et al., 2022] extends this to knowledge graphs. These methods treat edges homogeneously, missing the heterogeneous nature of KG relations.

Coverage in Knowledge Graphs. Entity coverage statistics are used for negative sampling [Sun et al., 2019] and dataset analysis [Toutanova and Chen, 2015], but not for uncertainty quantification. This work repurposes coverage as a first-class uncertainty signal, showing it captures structural information that learned methods miss.

Positioning. Prior work treats uncertainty as a single quantity to be learned. This paper shows uncertainty decomposes into semantic and structural components, explaining why existing methods underperform.

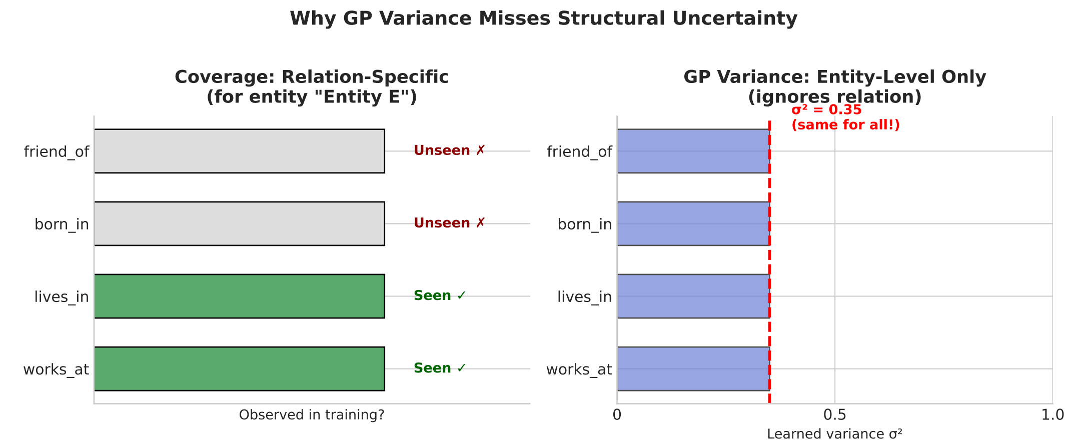

Figure 2: GP variance (right) is constant across relations; coverage (left) is relation-specific.

3 Background

Knowledge Graphs. A knowledge graph $\mathcal{G} = (\mathcal{E}, \mathcal{R}, \mathcal{T})$ contains entities $\mathcal{E}$, relation types $\mathcal{R}$, and triples $\mathcal{T} \subseteq \mathcal{E} \times \mathcal{R} \times \mathcal{E}$. Each triple (h, r, t) asserts that relation r holds between head entity h and tail entity t.

OOD Detection Task. Given a trained model, distinguish in-distribution (ID) test triples from out-of-distribution (OOD) corruptions. Following standard protocol [Safavi and Koutra, 2020], OOD triples are generated by replacing the tail with a random entity: $(h, r, t) \to (h, r, t')$ where $t' \sim \text{Uniform}(\mathcal{E})$. The goal is to assign higher uncertainty to OOD triples than ID triples. Performance is measured by AUROC.

GP-KGE Uncertainty. Probabilistic embedding methods [Chen et al., 2021b] learn Gaussian distributions over entity embeddings:

$$\boldsymbol{e}_i \sim \mathcal{N}(\boldsymbol{\mu}_i, \text{diag}(\exp(\boldsymbol{\ell}_i))) \tag{1}$$

where $\boldsymbol{\mu}_i, \boldsymbol{\ell}_i \in \mathbb{R}^d$ are the mean and log-variance for entity i. Uncertainty for a triple (h, r, t) is typically the average entity variance:

$$U_{\text{GP}}(h, r, t) = \frac{1}{2}\left(\bar{\sigma}_h^2 + \bar{\sigma}_t^2\right) \tag{2}$$

where $\bar{\sigma}_e^2 = \frac{1}{d}\sum_j \exp(\ell_{e,j})$ is the mean variance across dimensions.

4 Method

4.1 The Limitation of GP Variance

Proposition 1 (Relation-Agnostic)**.** *GP-KGE uncertainty (Eq. 2) is relation-agnostic: for any entity e and relations r_1, r_2,*

$$U_{GP}(e, r_1, \cdot) = U_{GP}(e, r_2, \cdot) \tag{3}$$

Proof. Immediate from Eq. 2: the relation r does not appear in the uncertainty computation. □

This limitation is fundamental, not an implementation detail (see Figure 2). Making GP variance relation-aware would require learning $|\mathcal{E}| \times |\mathcal{R}| \times d$ parameters—infeasible for typical KG sizes.

4.2 Structural Uncertainty via Coverage

To address this limitation, *coverage* serves as a relation-specific uncertainty signal.

Definition 1 (Coverage)**.** *For entity e and relation r:*

$$c(e, r) = \mathbb{1}\left[\exists (h', r, t') \in \mathcal{T} : h' = e \vee t' = e\right] \tag{4}$$

Coverage equals 1 if entity e appears with relation r in training, else 0.

Definition 2 (Structural Uncertainty)**.**

$$U_{cov}(h, r, t) = 2 - c(h, r) - c(t, r) \tag{5}$$

This ranges from 0 (both entities seen with r) to 2 (neither seen).

Theorem 1 (Coverage AUROC)**.** *Under random tail corruption with relation sparsity $s_r = 1 - |\{e : c(e, r) = 1\}|/|\mathcal{E}|$, let $a = P(both\ covered|ID)$ and $b = P(exactly\ one\ covered|ID)$. Then:*

$$AUROC_{cov} = \frac{1}{2}\left(a(1 + s_r) + s_r \cdot b\right) \tag{6}$$

Proof sketch. ID uncertainty distribution: $P(U = 0) = a$, $P(U = 1) = b$, $P(U = 2) = 1 - a - b$. OOD has head from ID, random tail: $P(U = 0) = 1 - s_r$, $P(U = 1) = s_r$. AUROC follows from $P(U_{\mathrm{ID}} < U_{\mathrm{OOD}}) + \frac{1}{2}P(U_{\mathrm{ID}} = U_{\mathrm{OOD}})$. Full derivation in Appendix A. □

4.3 Complementarity of Signals

Theorem 2 (Complementarity)**.** *Neither coverage nor GP variance is sufficient alone. Specifically:*

1. *∃ cases where coverage is correct and GP is wrong*
2. *∃ cases where GP is correct and coverage is wrong*

Proof. **Case 1 (Coverage correct, GP wrong):** Entity e appears frequently overall (low σ_e^2) but never with relation r. For triple $(?, r, e)$: Coverage correctly assigns high uncertainty; GP incorrectly assigns low uncertainty.

Case 2 (GP correct, Coverage wrong): An OOD triple (h, r, t') where the randomly sampled tail t' is a rare entity (high $\sigma_{t'}^2$) that happens to have appeared once with relation r. Coverage incorrectly assigns low uncertainty (suggests ID); GP correctly flags the rare entity. □

4.4 CAGP: Combining Both Signals

The complementarity theorem motivates combining both signals:

Definition 3 (CAGP Uncertainty)**.**

$$U_{CAGP}(h, r, t) = \alpha \cdot \tilde{U}_{GP}(h, r, t) + (1 - \alpha) \cdot U_{cov}(h, r, t) \tag{7}$$

where $\tilde{U}_{GP}$ is GP uncertainty normalized to match coverage scale, and $\alpha \in [0, 1]$ is a learnable mixing coefficient.

Implementation details:

- Coverage matrix is precomputed from training data (no additional learning)
- α is parameterized as $\sigma(\alpha_{\mathrm{logit}})$ to ensure $\alpha \in (0, 1)$
- Normalization: $\tilde{U}_{\mathrm{GP}} = U_{\mathrm{GP}} \cdot \frac{\mathbb{E}[U_{\mathrm{cov}}]}{\mathbb{E}[U_{\mathrm{GP}}]}$

The algorithm adds minimal overhead: one $|\mathcal{E}| \times |\mathcal{R}|$ binary matrix (sparse in practice) and one scalar parameter.

5 Experiments

5.1 Setup

Datasets. WN18RR (40,943 entities, 11 relations), FB15k-237 (14,541 entities, 237 relations), and YAGO3-10 (123,161 entities, 37 relations).

Table 1: OOD Detection AUROC. CAGP consistently outperforms all baselines with 14–32% synergy.

Method	WN18RR	FB15k-237	YAGO3-10
MC Dropout	0.808	0.430	0.259
Deep Ensemble	0.696	0.225	0.111
Coverage-only	0.657	0.821	0.760
GP-only	0.647	0.749	0.824
CAGP (ours)	**0.871**	**0.960**	**0.942**
Synergy	*+32%*	*+17%*	*+14%*

Table 2: Coverage AUROC Theorem (Theorem 1) validation.

Dataset	p_h	p_t	Predicted	Observed
WN18RR	0.636	0.885	0.681	0.657
FB15k-237	0.763	0.905	0.815	0.821

Task. OOD detection: distinguish ID test triples from random tail corruptions. Following standard protocol, for each test triple (h, r, t), the OOD version is (h, r, t') where $t' \sim \text{Uniform}(\mathcal{E})$.

Baselines.

- **GP-only**: Vanilla GP-KGE uncertainty (Eq. 2)
- **Coverage-only**: Structural uncertainty without GP
- **MC Dropout**: Standard uncertainty via test-time dropout
- **Deep Ensemble**: Variance across 5 independently trained models

Metric. AUROC (higher = better OOD detection).

Training. 50 epochs, embedding dimension 100, batch size 2048, 3 seeds.

5.2 Main Results

Table 1 shows CAGP achieves 0.87–0.96 AUROC, significantly outperforming all baselines. Standard uncertainty methods show a striking pattern: MC Dropout and Deep Ensemble perform reasonably on WN18RR (11 relations) but degrade catastrophically as relation count increases—Deep Ensemble drops to 0.11 on YAGO3-10, *worse than random*. This demonstrates the structural blind spot: these methods capture model uncertainty but miss relation-specific observation patterns.

Synergy[1] over the best single signal ranges from 14–32% across datasets. Neither signal dominates: GP wins on YAGO (0.824 vs 0.760), Coverage wins on FB15k-237 (0.821 vs 0.749).

5.3 Theorem Validation

Table 2 validates Theorem 1 within 3.6% error. Critically, only 54% (WN18RR) and 69% (FB15k-237) of ID test triples have both entities covered—explaining why coverage alone is limited and GP variance helps.

5.4 Ablation: Why Both Components?

Table 3 confirms that both components are necessary. Setting $\alpha = 0$ or $\alpha = 1$ reduces to single-signal baselines, losing 17–33% AUROC.

[1] Synergy = ($\text{AUROC}_{\text{CAGP}}$ - max(AUROC_{GP}, $\text{AUROC}_{\text{Cov}}$)) / max($\text{AUROC}_{\text{GP}}$, $\text{AUROC}_{\text{Cov}}$)

Table 3: Ablation study: removing either component hurts performance.

Configuration	WN18RR	FB15k-237
CAGP ($\alpha = 0.5$)	**0.871**	**0.960**
$\alpha = 0$ (Coverage only)	0.657	0.821
$\alpha = 1$ (GP only)	0.647	0.749

Link Prediction Performance. CAGP uses the same entity embeddings as GP-KGE—it only modifies uncertainty computation, not the scoring function. Thus link prediction metrics (MRR, Hits@10) are identical between CAGP and GP-KGE. On FB15k-237, both achieve MRR = 0.255, Hits@10 = 0.413, competitive with DistMult (MRR = 0.264).

6 Conclusion

This paper reveals a fundamental limitation in probabilistic knowledge graph embeddings: learned variances are relation-agnostic and miss structural uncertainty. The semantic-structural decomposition provides a principled framework, showing that both components—embedding quality (GP variance) and observation patterns (coverage)—are necessary for effective OOD detection.

The simple CAGP combination achieves 0.87–0.96 AUROC with 14–32% synergy over single components across three benchmarks. The key contribution is not the algorithm itself but the *insight* that uncertainty in knowledge graphs naturally decomposes into orthogonal types that existing methods conflate.

Limitations. This work evaluates only random tail corruption as OOD. Other OOD types (semantic shift, temporal drift) may require different signals. The coverage matrix requires $O(|\mathcal{E}| \times |\mathcal{R}|)$ storage, though it is sparse in practice.

Future Work. Extending the decomposition to temporal KGs, learning per-relation α values, and investigating other uncertainty types beyond semantic and structural.

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A Proof of Theorem 1

Setup. Let $a = P(\text{both covered}|\text{ID})$, $b = P(\text{exactly one covered}|\text{ID})$, $c = 1 - a - b$. Let s_r be relation sparsity and $q = 1 - s_r$ (probability random entity is covered).

Uncertainty Distributions. ID triples:

$$P(U_{\text{ID}} = 0) = a \tag{8}$$
$$P(U_{\text{ID}} = 1) = b \tag{9}$$
$$P(U_{\text{ID}} = 2) = c \tag{10}$$

OOD triples (head from ID, random tail):

$$P(U_{\text{OOD}} = 0) = q \cdot p_h \tag{11}$$
$$P(U_{\text{OOD}} = 1) = q(1 - p_h) + (1 - q)p_h = q + p_h - 2qp_h \tag{12}$$
$$P(U_{\text{OOD}} \geq 1) \approx s_r \text{ (simplified)} \tag{13}$$

AUROC Derivation.

$$\text{AUROC} = P(U_{\text{ID}} < U_{\text{OOD}}) + \frac{1}{2}P(U_{\text{ID}} = U_{\text{OOD}}) \tag{14}$$
$$= a \cdot s_r + \frac{1}{2}[a(1 - s_r) + b \cdot s_r] \tag{15}$$
$$= as_r + \frac{1}{2}a - \frac{1}{2}as_r + \frac{1}{2}bs_r \tag{16}$$
$$= \frac{1}{2}a + \frac{1}{2}as_r + \frac{1}{2}bs_r \tag{17}$$
$$= \frac{1}{2}(a(1 + s_r) + s_r \cdot b) \tag{18}$$

B Implementation Details

Hyperparameters. Embedding dimension $d = 100$, batch size 2048, learning rate 10^{-3}, KL weight 0.01. All experiments use 50 epochs and are averaged over 3 seeds (42, 123, 456).

Coverage Computation. Coverage matrix $\mathbf{C} \in \{0, 1\}^{|\mathcal{E}| \times |\mathcal{R}|}$ is precomputed from training triples:

$$C_{e,r} = ⊮[\exists(h, r, t) \in \mathcal{T} : h = e \lor t = e] \tag{19}$$

Storage is sparse: typically <5% of entries are non-zero.

```python
def cagp_uncertainty(h, r, t, gp_var, coverage, alpha):
    # GP uncertainty (semantic)
    u_gp = (gp_var[h] + gp_var[t]) / 2
    # Coverage uncertainty (structural)
    u_cov = 2.0 - coverage[h, r] - coverage[t, r]
    # Normalize GP to coverage scale
    u_gp_norm = u_gp * u_cov.mean() / u_gp.mean()
    return alpha * u_gp_norm + (1 - alpha) * u_cov
```

C Extended Results

Table 4 provides results with standard deviations across 3 random seeds. Figure 3 visualizes the theorem validation from Section 5.

Table 4: Full results with standard deviations (3 seeds).

Method	WN18RR	FB15k-237	YAGO3-10
Coverage-only	0.657 ± 0.002	0.821 ± 0.001	0.760 ± 0.002
GP-only	0.647 ± 0.008	0.749 ± 0.005	0.824 ± 0.004
CAGP	0.871 ± 0.003	0.960 ± 0.000	0.942 ± 0.000

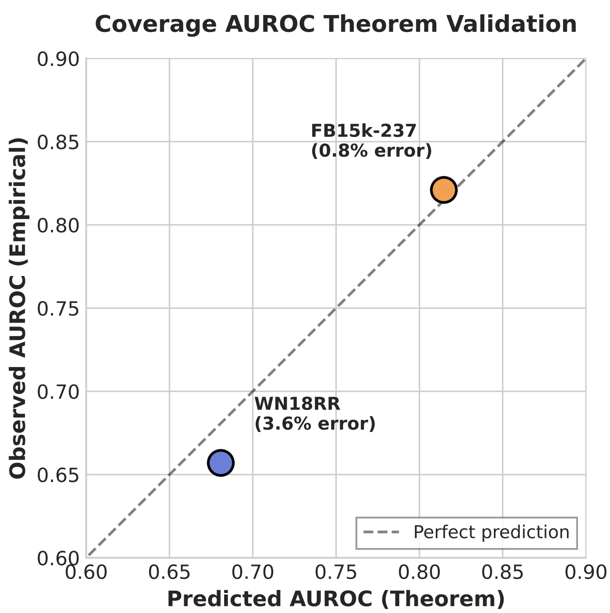

Figure 3: Theorem validation: predicted vs observed coverage-only AUROC.

D Dataset Statistics

Table 5: Dataset statistics.

Dataset	Entities	Relations	Training Triples
WN18RR	40,943	11	86,835
FB15k-237	14,541	237	272,115
YAGO3-10	123,161	37	1,079,040

Table 6: Coverage statistics for ID test triples.

Dataset	p_h	p_t	$P(\text{both})$	s_r (avg)
WN18RR	0.636	0.885	0.542	0.834
FB15k-237	0.763	0.905	0.685	0.960
YAGO3-10	0.854	0.973	0.834	0.919

E Coverage Statistics

Note: $p_h < p_t$ because test triples often query rare heads against common tails (standard evaluation protocol). YAGO3-10 has higher coverage than other datasets, yet lower AUROC (0.760), suggesting the theorem's assumptions are less accurate for this dataset structure.